

Module 04: Cognition — Neural Substrates of Belief Updating

Course: Active Inference 101 | **Unit:** Computational Neuroscience | **Audience:** First-semester undergraduates

Learning Objectives

1. Describe how the **prefrontal cortex** maintains and updates beliefs through working memory.
2. Explain how **attention** is implemented neurally through precision-weighted gain modulation.
3. Identify neural correlates of free energy minimization in working memory and decision-making circuits.

Introduction

Cognition — thinking, reasoning, deciding — depends on the brain's ability to maintain beliefs, update them with new evidence, and direct attention to relevant information. This module explores the neural circuits that make cognition possible.

Key Concepts

1. Prefrontal Cortex and Working Memory

The **prefrontal cortex (PFC)** is the brain's executive hub. It maintains **working memory** — the ability to hold information in mind temporarily:

- PFC neurons can sustain activity for seconds, maintaining a representation (belief) even after the stimulus is gone
- This persistent activity represents the brain's current "best guess" about the relevant hidden states

- Different PFC regions handle different aspects: dorsolateral PFC for spatial/abstract, ventrolateral PFC for object/verbal

In Active Inference, working memory is the neural substrate of the approximate posterior $q(s)$ — the brain's current beliefs about hidden states.

2. Attention Networks

The brain has two major attention networks:

- **Dorsal attention network** (top-down): Voluntary, goal-directed attention. "I will focus on the lecture." Increases precision on task-relevant information.
- **Ventral attention network** (bottom-up): Stimulus-driven, involuntary attention. "What was that sound?" Detects salient (high prediction error) stimuli.

Both implement precision weighting — the dorsal network turns up the gain on what you *want* to attend to, while the ventral network automatically boosts signals that are unexpectedly large.

3. Prefrontal-Parietal Interactions

Cognition involves tight collaboration between the PFC and parietal cortex:

- PFC generates abstract goals and maintains context
- Parietal cortex represents spatial relationships and current sensory state
- Together, they form a **fronto-parietal network** that implements goal-directed behavior

When this network is disrupted (e.g., in ADHD, schizophrenia), the ability to maintain context, shift attention, and update beliefs is impaired.

4. Neural Correlates of Belief Updating

When the brain updates beliefs, specific neural signatures can be measured:

- **P300 event-related potential**: A positive voltage deflection ~300ms after a surprising stimulus — reflects the updating of context/beliefs

- **Frontal theta oscillations (4-8 Hz):** Increase during cognitive control, conflict detection, and model updating
- **Reduced prefrontal activity with learning:** As a situation becomes predictable, PFC activity decreases — less updating is needed

5. Computational Psychiatry

Disruptions in belief updating circuits produce characteristic psychiatric symptoms:

- **Schizophrenia:** Aberrant precision on prediction errors → inappropriate belief updating → delusions
- **ADHD:** Impaired precision modulation → difficulty maintaining focus and filtering irrelevant signals
- **Depression:** Reduced gain on positive prediction errors → difficulty updating toward positive beliefs
- **Anxiety:** Excessive gain on threat-related prediction errors → overestimation of danger

Summary

Cognition is implemented through prefrontal working memory circuits that maintain beliefs, attention networks that control precision weighting, and fronto-parietal interactions that coordinate goal-directed behavior. Neural signatures like the P300 and frontal theta reveal real-time belief updating. Psychiatric conditions can be understood as characteristic disruptions of these precision-weighting mechanisms.

Further Reading

- Miller, E. K. & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167-202.
- Corbetta, M. & Shulman, G. L. (2002). Control of goal-directed and stimulus-driven attention in the brain. *Nature Reviews Neuroscience*, 3(3), 201-215.
- Corlett, P. R. et al. (2019). Hallucinations and strong priors. *Trends in Cognitive Sciences*, 23(2), 114-127.